

Derived or Observed: A Duty Written From First Principles Covers the Hazards It Can Derive and Misses the Ones Only Observation Reveals A Reproducible Single-Turn Assay of Procedural-Specification Format and Hazard Coverage

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Abstract

A procedure can be written in more than one form. We compare two forms of a *duty* (a gated procedure that governs a class of work) for repairing a confirmed software defect. The *annotated* form is written by reading a hazard registry and citing each entry it guards against, so the finished procedure depends on that registry to be understood. The *cartographer* form is written from first-principles analysis of the task, cites nothing, and is self-sufficient: an executor needs no external reference to follow it. We ask which hazards each form's procedure actually prevents. A local open-weight model (Qwen3.8-27B) writes the procedure in a single completion over a bare inference endpoint, and a blind judge scores, for each of ten pre-registered hazards, whether the procedure contains a step that would prevent it. The hazards are split in advance into seven that are derivable from the task and three that are not: a routing obligation upstream of the task, a session-close recording obligation, and a durability distinction learnable only from accumulated observation. Given the registry and told to write one gate per entry, the annotated form covers every hazard (210 of 210 derivable, 90 of 90 non-derivable); this is a manipulation check that the model follows the method, not a discovered effect. The two informative contrasts are between the first-principles conditions. The cartographer form covers the derivable hazards at a per-duty rate indistinguishable from a model given no method at all (0.64 versus 0.60; permutation $p=0.44$): writing from first principles buys self-sufficiency, not coverage. And it covers the non-derivable hazards almost never: 1 of 30 duties reaches any of them, against 30 of 30 for the annotated form (Fisher $p=5\times 10^{-16}$). The pre-registered condition-by-class interaction, tested with the duty as the unit, is decisive (permutation $p<10^{-4}$). The cartographer form's distinctive property is self-sufficiency: it cites no registry entry in any run, where the annotated form cites all ten in every run. We then run a fix the form's proposers specified but never executed: a meta-taboo scan that names two hazard classes, routing and session-close. It recovers exactly those two (routing 0 to 27 of 30; session-close 1 to 25 of 30) and not the third, the pure observation-only durability hazard it does not name (0 to 0 of 30). A blind rater judged coverage and an independent rater agreed (Cohen's $\kappa=0.85$); the direction is also corroborated without any judge, as the cartographer form never mentions routing across 30 runs and the scan form always does. Naming a class of hazard lets first-principles derivation reach it; a hazard whose badness is only apparent from accumulated observation, and that belongs to no named class, stays out of reach. The finding is a direction-and-rate result on one defect specimen and one model; the materials, runs, and scoring are public and reproduce on a 24 GB GPU with no paid API.

1 Introduction

Two people can write the same procedure in different forms. One writes by consulting a registry of known hazards and citing each one the procedure guards against; the reader of that procedure must know the registry to understand it. The other writes from first principles, analyzing the task and encoding each step so it stands on its own; the reader needs nothing but the procedure. The second form is more portable, since it carries its own justification, and a research program that writes machine-followed procedures (*duties*) prefers it for that reason. This paper asks what the portable form costs. A procedure written without the registry can only guard against the hazards its author can derive. If some hazards are not derivable from the task, if their danger is apparent only from having seen them happen, the self-sufficient form has no way to reach them.

We make this concrete with a repair-duty: a duty governing how an office repairs a confirmed software defect at a known locus. We hand a local open-weight model one of two writing methods and a repair task, and it emits the repair-duty in a single completion. We then score, for each of ten pre-registered hazards, whether the produced duty contains a gate that would prevent it. The ten hazards are pre-split into those derivable from the repair act and those that are not. The question is whether the form of the writing method decides which hazards the produced duty covers, and whether that decision falls along the derivable/not-derivable line.

This reformulates an earlier two-run study in the same program, which reported the pattern from a single pair of hand-written procedures and proposed but never ran a fix. We restate it as a quantified, pre-registered, reproducible assay and run the fix.

1.1 Related Work

Whether the structure of a written specification changes how a model follows it is an active question for coding agents. Gloaguen et al. (2026) evaluate repository-level context files and find that an imperative instruction is followed while descriptive repository overviews add little. McMillan (2026) runs a factorial study of file-structure variables in coding-agent configuration files. Geng et al. (2026) measure the pragmatic influence of instruction wording, and Pecher et al. (2026) show classification behavior shifts with prompt underspecification. Our contribution is narrower and orthogonal to these: we hold the task fixed and vary only whether the procedure-writing method reads a hazard registry, and we measure hazard coverage of the produced procedure rather than task accuracy. We searched Google Scholar and arXiv (September 2026) for work on whether the form of a procedural specification determines which hazards it covers, as distinct from whether it is followed; we found none directly on point, and the coding-agent configuration studies above are the nearest neighbors. The duty, taboo, and cartographer/annotated vocabulary is internal to this research program; the reformulated study it draws on is Neilson (in preparation).

2 Materials and Method

2.1 The reformulation

The earlier study used a multi-turn agent that read files and wrote a procedure to disk. That subject cannot run on the reproducible endpoint this program requires, so the assay is recast as a single-turn generation: the model receives a writing method and the repair task, and emits the complete repair-duty as its one reply. There is no harness, no tool use, and no second turn; the produced text is the artifact scored. This preserves the question (does the writing method’s form decide hazard coverage?) while fitting an endpoint anyone can reproduce.

2.2 Conditions and materials

The independent variable is the writing method. Information is not matched across conditions, by design: the two forms are defined by their relationship to the hazard registry, so the annotated method carries the registry and the cartographer method does not. This asymmetry is the object of study, not a confound. A hazard whose danger is only apparent from accumulated observation cannot be derived from the task; withholding the registry from the cartographer condition is what makes “does first-principles derivation reach this hazard?” a real question rather than “will the model copy a list it can see?” A pilot with the registry placed in the shared prompt confirmed the latter: the model copied the observation-only hazards and the effect vanished.

- **baseline**: the repair task and the defect only, no writing method and no registry. The floor.
- **annotated**: the annotated method, carrying the ten-hazard registry: read it, open with a hazard table, write one gate per hazard, and cite the registry entry in each gate body.
- **cartographer**: the cartographer method, no registry: derive the failure modes from the repair act, encode each as a self-sufficient gate, cite nothing.
- **cartographer+scan**: the cartographer method plus a meta-taboo scan that names two hazard classes, routing (upstream of the act) and session-close obligations (after the act), and directs a structural gate for each.

The shared task names one defect: `calculate_average` divides by `len(values)` and raises on an empty list, locus confirmed. The writing methods and the registry are reproduced verbatim in Appendix A.

2.3 The ten hazards and the pre-registered split

Each hazard names a divergence point a repair can cross. Seven are derivable from analyzing the repair act (a confirmed locus before editing; a declared pass criterion; a scope boundary; fixing the canonical source not a call site; a revised root cause before a second attempt; recorded completion evidence; verification by observed result not claim). Three are not: a *routing* obligation upstream of the act (confirm a repair, not another task, is what is needed); a *session-close* obligation (record a constraint discovered during the repair before closing); and a *durability* distinction (advisory versus structural fixes), whose badness is learnable only from accumulated observation. The split, the divergence points, and which hazards the scan names are fixed before the run in `studies/cartographer-duty-format/taboo_set.json`.

2.4 Model and Sampling

The subject is Qwen3.8-27B (Q4_K_M GGUF) served over a bare `llama.cpp` raw completion endpoint (no chat template, no tools) with thinking pinned off in the published wrapper. The complete sampler chain is declared in the study definition and sent with every request: temperature 0.8; `min_p` 0.05 as the sole live truncation stage; `top_k`, `top_p`, `typical_p`, `top_n_sigma`, all penalties, and the XTC and DRY stages off; one seed per replicate (1 through 30); a 4096-token cap. Each condition ran 30 replicates. The server ran on a GPU (RTX 3090); recorded throughput held between 37.9 and 40.8 tokens per second across all 120 runs, with no step down to a CPU rate, and no response was truncated at the cap (the longest was 3806 predicted tokens). The served build, read from the endpoint’s `props` at the session, was `b9445-af6528e6d`; the per-row build field was empty in this run’s record, a recorded gap noted here rather than inferred. The probe image digest and the substrate probe are in the run record. The repository was on commit `56eb443` with a dirty working tree when the run launched, disclosed here; the committed materials and instrument are at the follow-up commit named in Data Availability.

2.5 Scoring

Two measurements. The first is deterministic: a script counts how many of the ten registry slugs each produced duty cites (`slug_c2.py`), which needs no judge. The second is coverage: for each duty and each hazard, does the duty contain a gate whose action would prevent that hazard’s divergence point? Naming a hazard is not coverage; the gate’s action is. Coverage was judged by Claude (Opus 4.8), blind to condition: the 120 duties were shuffled and stripped of their condition labels into an opaque-id set before judging, and unblinded only for analysis. An independent second rater, also blind and given a clean rubric with no worked example, re-scored a stratified 40-duty sample; agreement is reported in Section 3.

2.6 Agents and Models

Role	Model	Configuration	Date
Subject	Qwen3.8-27B	Q4_K_M GGUF, llama.cpp build b9445-af6528e6d (per props), raw completion, thinking off, GPU (RTX 3090), context 32768.	2026-09-11
Slug scorer (C2)	Deterministic rule	<code>slug_c2.py</code> ; exact registry-slug match on the response text; no judgment.	2026-09-11
Coverage judge	Claude Opus 4.8	Blind to condition (shuffled, de-labelled set); four batches over the 120 duties. Not a treatment arm.	2026-09-12
Second rater	Claude Opus 4.8	Independent session, blind, clean rubric with no worked example; stratified 40-duty sample. Same family as the primary judge (see Limitations).	2026-09-12
Author and operator	Claude Opus 4.8	Authored the materials, ran the study, wrote the coverage rubric. Not a treatment subject.	2026-09-12

Table 1: Agents and models. The subject is the only treatment arm and is a local open-weight model; Claude is used only to judge and to operate, never as a treatment condition.

3 Results

3.1 Slug citation is a clean, deterministic separation

The annotated method cites all ten registry slugs in every one of its 30 runs (mean 10.0 slugs). The baseline, cartographer, and cartographer+scan conditions cite none, in any run. The self-sufficiency the cartographer form is written for holds exactly: its produced duties carry no dependency on the registry, where the annotated form’s carry all of it.

3.2 Coverage splits along the derivable line

Table 2 gives coverage by hazard class as covered/judged calls, and the per-duty means used for the tests below. The annotated method covers everything (210 of 210, 90 of 90). This is a manipulation check, not a discovered effect: the annotated method is handed the exact ten-hazard registry and told to write one gate per entry, so full coverage is a ceiling by construction, and it confirms only that the model followed the method. The scientific weight is in the two contrasts among the first-principles conditions.

All inferential tests below take the duty as the unit of analysis ($n=30$ per condition), because the 3 or 7 hazard calls within one produced duty are not independent; the covered/judged counts in the table are reported as descriptive rates, not as independent trials.

First, the cartographer method’s derivable coverage is not distinguishable from the no-method baseline: per-duty rate 0.64 versus 0.60 (permutation test on per-duty rates, $p=0.44$). Writing from first principles does not raise coverage over giving the model no method at all; what it changes is the dependency, not the coverage. This is a failure to reject, not a demonstration of equivalence: the per-duty rates overlap and a small real difference is not excluded. Second, the two methods diverge almost completely on the non-derivable hazards: every one of the 30 annotated duties reaches all three, and only 1 of the 30 cartographer duties reaches any (Fisher exact on duties, $p=5\times 10^{-16}$). The pre-registered condition-by-class interaction is the difference between these: the annotated advantage over cartographer is 0.36 on derivable hazards and 0.99 on non-derivable ones. Tested at the duty level as the difference-in-differences (each duty’s derivable rate minus its non-derivable rate, annotated versus cartographer), the interaction is decisive: the annotated per-duty difference is 0.00 and the cartographer’s is +0.63 (permutation test, $p<10^{-4}$; no resample of 100,000 reached the observed separation).

Within the derivable set the cartographer method is heterogeneous, so the pooled 0.64 flattens it: per-taboo coverage runs from 2 of 30 (revising the root cause before a second attempt) to 30 of 30 (requiring a confirmed locus), and two derivable hazards are missed badly (Section 3.5).

Condition	Derivable (7 hazards)	Non-derivable (3 hazards)
baseline	127/210 (0.60)	4/90 (0.04)
annotated	210/210 (1.00)	90/90 (1.00)
cartographer	134/210 (0.64)	1/90 (0.01)
cartographer+scan	133/210 (0.63)	52/90 (0.58)

Table 2: Hazard coverage by class, as covered/judged calls with the per-duty mean rate in parentheses, over 30 duties per condition and 7 or 3 hazards. Derivable hazards are those a first-principles analysis of the repair act can reach; non-derivable are the routing, session-close, and durability hazards. The annotated row is a construction ceiling (see text); inferential tests use the per-duty rates.

3.3 The scan recovers the classes it names

The meta-taboo scan names two hazard classes. Table 3 shows it recovers exactly those two and not the third. Routing coverage rises from 0 of 30 to 27 of 30, and session-close from 1 of 30 to 25 of 30. The durability hazard, which the scan does not name and which is learnable only from observation, stays at 0 of 30. Naming a class of hazard lets first-principles derivation reach it; the hazard that belongs to no named class remains unreachable.

Hazard	Class	Scan names it	cartographer	+scan	Fisher p
routing	meta/routing	yes	0/30	27/30	9.2×10^{-14}
session-close	session-close	yes	1/30	25/30	1.2×10^{-10}
durability	observation-only	no	0/30	0/30	1.0

Table 3: Cartographer versus cartographer+scan on the three non-derivable hazards. The scan recovers the two classes it names and not the one it does not.

3.4 The direction holds without a judge

Coverage is a judged measurement, so we corroborate its direction with a deterministic keyword-presence check that needs no rater: does a produced duty even mention the subject of each non-derivable hazard? The pattern matches the judged coverage. No cartographer duty mentions routing (0 of 30) and every scan duty does (30 of 30), tracking the judged 0-to-27 recovery. The advisory or durability vocabulary appears in 2 of 30 cartographer duties and 3 of 30 scan duties, tracking the judged non-recovery of the durability hazard. The annotated duties mention all three subjects in every run (30 of 30 each). Mentioning a subject is weaker than covering it with a gate, so these counts sit at or above the judged coverage rates; their value is that the divergence between the forms is visible in the raw text without any judgment.

3.5 Two nominally derivable hazards behave like the hard ones

The derivable/not-derivable split predicts the broad pattern, but it is not perfectly clean. Two hazards labelled derivable are missed by the cartographer method almost as often as the non-derivable ones: fixing the canonical source rather than a call site (7 of 30) and revising the root cause before a second attempt (2 of 30). Only the annotated method, carrying the registry, covers them reliably (30 of 30 each). These two sit closer to the hard end than their label implies: they are derivable in principle but subtle enough that a from-scratch analysis usually does not surface them. We report the split as pre-registered and flag these two rather than re-labelling them after the fact.

3.6 Inter-rater agreement

The independent second rater, blind and given a clean rubric with no worked example, re-scored a stratified 40-duty sample (10 per condition). Agreement with the primary judge was 372 of 400 hazard calls (0.93; Cohen's $\kappa = 0.85$). The disagreements concentrate on one borderline hazard, whether recorded completion evidence is required (29 of 40); the hazards that carry the finding agree strongly, including the observation-only durability hazard at 40 of 40 and routing at 39 of 40. Because the second rater saw no worked example, the agreement indicates the primary coverage calls are not an artifact of the worked example in the primary rubric.

4 Discussion

The result draws a line under a portable procedure. A duty written from first principles, needing no external reference, is exactly as good as its author's ability to derive hazards from the task, and no better, on this evidence, than a model given no method at all. Its value is that it carries no dependency, not that it prevents more. The hazards it misses are the ones a task cannot teach: an upstream routing check, a session-close obligation, and a durability distinction that only accumulated observation reveals. The annotated form reaches all of them because it copies them from a registry that already remembers them.

The scan result sharpens this. A hazard that a first-principles author misses is not always unreachable; it is unreachable until it is named. Naming the class ("check for routing hazards," "check for session-close obligations") gives the derivation a place to look, and coverage of those classes jumps from near zero to most runs. What naming does not rescue is the hazard whose content, not just its existence, must be observed: the advisory-versus-structural durability distinction stays at zero even with the scan running, because knowing to look for durability hazards does not tell you what makes an advisory fix fragile. That is the residue that a self-sufficient format cannot, in principle, carry.

4.1 Limitations

One defect specimen, one model, single-turn generation. This is a direction-and-rate finding on the `calculate_average` empty-list defect as written for Qwen3.8-27B; the derivable/not-derivable boundary may fall differently for another domain, another model, or a multi-turn agent that can consult a registry by choice rather than having it withheld. The information asymmetry between the forms is deliberate and is discussed in the Method; a consequence is that the non-derivable coverage gap is partly expected, since a hazard that must be observed cannot be derived, so the informative quantities are how much of the derivable set the cartographer form recovers (most) and whether naming a class recovers the rest (for two of three, yes).

The coverage result is judge-dependent, and both the coverage judge and the second rater are Claude (Opus 4.8). This is a real limit: two instances of one architecture can agree because they share a blind spot, so the $\kappa=0.85$ is within-family agreement exposed to correlated error, not cross-architecture confirmation. What does not depend on a judge is narrower than the whole finding: the deterministic slug count establishes only the self-sufficiency of the cartographer form (it cites no registry entry), not the coverage rates. The coverage rates (the 0.64, the 0.01, the scan recovery) rest on the Claude judge. Two things bound the risk. The primary judge saw a worked example in its rubric that could have anchored it; the second rater was given no such example, and the two agree, so the calls are not an artifact of that example. And the direction is corroborated by the rater-free keyword-presence check (Section 3): the routing gap and the durability non-recovery are visible in the raw text with no judgment at all. A local cross-architecture coverage rater remains future work.

Two smaller items. The subject is named by its served GGUF alias, `Qwen3.8-27B-Q4_K_M.gguf`; the endpoint attests that filename, not an upstream release identity, so the model claim is only as good as the local artifact name. And of the earlier study's added criteria, C5 (a coverage-map slug leak in the cartographer output) is answered by the slug count: no cartographer or scan duty cited any slug (0 of 30 each), so there is no leak. C8 (a patchover check for proxy completion conditions) requires a separate judgment we did not perform and is retired here, as C7 was retired as not applicable to single-turn generation.

The pre-registered derivable split is imperfect for two hazards, disclosed above and not re-labelled.

5 Conclusion

A self-sufficient procedure covers the hazards its author can derive from the task and misses the ones that can only be observed. Writing from first principles does not, on this evidence, prevent more failures than no method at all; it removes the dependency on a registry, and with it the reach to any hazard the registry alone remembers. Naming a class of hazard restores the reach for hazards whose existence is the only thing that must be supplied; it does not restore reach for a hazard whose content must be observed. A portable format and complete hazard coverage are, in this setting, two different things.

5.1 Data Availability

The complete study, comprising the writing methods, the shared task and defect, the pre-registered hazard set and split, the design and pre-registration documents, the run records with per-row provenance, the produced duties, the blind judging set and map, the deterministic slug scorer, the coverage grid, and the analysis script, is available at <https://github.com/sophdn/corpos-lab> under the `studies/cartographer-duty-format/` directory. The instrument change that added the three format conditions is in the same repository under `internal/`.

References

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A Materials

The writing methods (annotated, cartographer, cartographer+scan), the shared scenario with the defect, and the ten-hazard registry are reproduced verbatim in the study directory named in Data Availability (`studies/cartographer-duty-format/materials/`). The pre-registered hazard split and the scan-target flags are in the same directory’s `taboo_set.json`, and the coverage judging brief is in `scores/JUDGE_BRIEF.md`.